

Chemistry 30 — Unit III

Electrochemical Changes – Lesson Notes

Lesson 2 — Balancing Redox Equations using Oxidation Numbers and Redox Titrations

What you will be able to do

By the end of this lesson, you will be able to write and balance oxidation–reduction (redox) equations in acidic and neutral solutions by assigning oxidation numbers and by using half-reaction methods. You will learn how to build half-reactions from redox information and how to combine them to produce a correctly balanced overall equation.

You will also apply balanced redox equations to quantitative chemistry by performing calculations involving redox titrations. These skills allow chemists to determine unknown concentrations in real samples (for example, hydrogen peroxide in disinfectants, iron(II) in water samples, or oxidizing agents in bleaches).

By the End of This Lesson, You Will Be Able To:

KEY IDEA

- ✓ **Assign Oxidation Numbers**
Determine oxidation states of all elements, 0 for elemental forms.
- ✓ Use known oxidation states and by using half-reaction methods.
- ✓ You will learn how to build half-reactions from redox information.

Mn +7

Write and Balance Redox Equations

$\text{MnO}_4^- + \text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + \text{Fe}^{3+}$

$\text{O}_2^- = -2$
 $\text{H}^+ = +1$

$\text{Zn} + \text{VO}_2^+ + 4\text{H}^+ \rightarrow \text{Zn}^{2+} + \text{VO}_2^+ + \text{H}_2\text{O}$

$\text{Zn} + \text{VO}_2^+ + 4\text{H}^+ \rightarrow \text{Zn}^{2+} + \text{VO}_2^+ + 2\text{H}_2\text{O}$

1 Prepare Solutions

- ✓ Titrant in burette, analyte in flask.
- ✓ Add indicator to analyte (if needed).

2 Perform Titration

- ✓ Slowly add titrant to analyte.
- ✓ Swirl flask constantly.

3 Determine Result

- ✓ Measure final burette volume.
- ✓ Calculate moles of titrant used.
- ✓ Use mole ratio to find moles of analyte.
- ✓ Calculate analyte concentration.

$$\frac{0.0200 \text{ mol KMnO}_4}{\text{L Fe}^{2+}} = \frac{0.0224 \text{ L KMnO}_4}{\text{L Fe}^{2+}} = 0.0896$$

$$\frac{0.02224}{0.0250} = 0.0896 \text{ M Fe}^{2+}$$

PART A — A Systematic Strategy for Balancing Redox Equations

In Lesson 1 you learned how to identify oxidation and reduction using oxidation numbers and how to identify oxidizing and reducing agents. In this lesson, you will learn a reliable step-by-step method for balancing redox equations in aqueous solutions. The goal is to produce equations that conserve both atoms and electric charge.

Why Balancing Redox Equations is Different

In many reaction types, there is an additional requirement:

- ✓ Mass must be conserved (the number of each type of atom is the same on both sides).
- ✓ Charge must be conserved (electrons are transferred, not created or destroyed).

Core idea: electrons must cancel

A redox equation is correctly balanced only when:

- ✓ The oxidation half-reaction shows electrons as products (electrons are lost).
- ✓ The reduction half-reaction shows electrons as reactants (electrons are gained).

Step-by-step overview:

Use this sequence every time. With practice, it becomes automatic:

1 Skeleton equation

- ✓ Assign oxidation numbers. $\rightarrow \text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$
- ✓ Split into half-reactions. $\rightarrow \text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightarrow \text{VO}_2^+$



Step-by-step overview (half-reaction method):

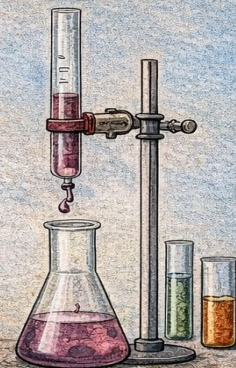
Use this sequence every time. With practice, it becomes automatic.

2 Perform Titrations

- ✓ Measure final burette volume,
- ✓ Calculate moles of titrant used,
- ✓ Use mole ratio to find moles of analyte.



$$\frac{0.00224}{\text{L Fe}^{2+}} / \frac{0.0250}{\text{M Fe}^{2+}}$$



Why balancing redox equations is different

In many reaction types (for example, synthesis or double replacement), balancing an equation is mainly an atom-counting problem. In redox reactions, there is an additional requirement:

- Mass must be conserved (the number of each type of atom is the same on both sides).
- Charge must be conserved (electrons are transferred, not created or destroyed).

Because redox reactions involve electron transfer, a balanced redox equation must correctly account for how many electrons are lost and how many are gained.

Two common approaches

In Chemistry 30, you will balance redox equations using two closely related tools:

- 1) Oxidation numbers — to identify what is oxidized/reduced and to determine how many electrons are

transferred.

2) Half-reactions — to show oxidation and reduction as separate processes, then recombine them into one overall balanced equation.

Throughout this unit, you will often begin with oxidation numbers (to understand the electron changes) and then use half-reactions to finish a fully balanced equation.

Before you start: identify the reaction environment

Balancing steps depend on whether the reaction occurs in:

- Acidic solution: $\text{H}^+(\text{aq})$ and $\text{H}_2\text{O}(\text{l})$ are allowed for balancing hydrogen and oxygen.
- Neutral solution: $\text{H}_2\text{O}(\text{l})$ is allowed; $\text{H}^+(\text{aq})$ is not left in the final answer.

In this course, we will primarily practice acidic and neutral solutions (not strongly basic). If H^+ is used during balancing in neutral solution, it must be removed from the final equation.

Core idea: electrons must cancel

A redox equation is correctly balanced only when:

- The oxidation half-reaction shows electrons as products (electrons are lost).
- The reduction half-reaction shows electrons as reactants (electrons are gained).
- The number of electrons lost equals the number gained, so electrons cancel when the half-reactions are added.

Step-by-step overview (half-reaction method)

Use this sequence every time. With practice, it becomes automatic:

Step 1 — Write the skeleton equation.

Write correct formulas for reactants and products. If species are ionic, show charges.

Step 2 — Assign oxidation numbers.

Determine which element is oxidized (oxidation number increases) and which is reduced (oxidation number decreases).

Step 3 — Split into half-reactions.

Write one half-reaction for oxidation and one for reduction using only the species that change.

Step 4 — Balance atoms other than H and O.

Balance all elements except hydrogen and oxygen first.

Step 5 — Balance oxygen using H₂O.

Add H₂O(l) to the side that needs oxygen atoms.

Step 6 — Balance hydrogen.

- In acidic solution: add H⁺(aq) to the side that needs hydrogen.
- In neutral solution: you may temporarily use H⁺(aq) to balance, but it must be removed later.

Step 7 — Balance charge using electrons.

Add e⁻ to the more positive side of each half-reaction until total charge is equal on both sides.

Step 8 — Make electrons equal and add the half-reactions.

Multiply one or both half-reactions so the number of electrons lost equals the number gained.

Add the half-reactions and cancel identical species (including e⁻).

Step 9 — Final clean-up.

Cancel any H₂O, H⁺, or other species that appear on both sides. Confirm:

- atoms balanced
- net charge balanced
- coefficients are in lowest whole-number ratio

A quick accuracy check

After balancing, do two checks:

- 1) Atom check: count each element on both sides.
- 2) Charge check: add the charges on each side (treating coefficients correctly). The totals must match.

In the next part, you will apply this method to full examples in acidic and neutral solutions.

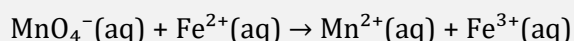
PART B — Balancing Redox Equations in Acidic Solution (Worked Example)

The most effective way to learn the half-reaction method is to see it applied carefully, step by step. In this part, you will work through a complete example of balancing a redox equation that occurs in acidic solution.

Example: $\text{MnO}_4^- (\text{aq}) + \text{Fe}^{2+} (\text{aq}) \rightarrow \text{Fe}^{3+} (\text{aq}) + \text{Mn}^{2+} (\text{aq})$

- 1 Write the skeleton equation**
 $\text{MnO}_4^- (\text{aq}) + \text{Fe}^{2+} (\text{aq}) \rightarrow \text{Mn}^{2+} (\text{aq}) + \text{Fe}^{3+} (\text{aq})$
- 2 Assign oxidation numbers**
 $\text{MnO}_4^- (\text{aq}) + \text{Fe}^{2+} (\text{aq}) \rightarrow \text{Mn}^{2+} (\text{aq}) + \text{Fe}^{3+} (\text{aq})$
Reduction half-reaction: $\text{MnO}_4^- \rightarrow \text{Mn}^{2+} (\text{aq})$
Oxidation half-reaction: $\text{Fe}^{2+} (\text{aq}) \rightarrow \text{Fe}^{3+} (\text{aq})$
- 3 Write half-reactions**
Reduction half-reaction: $\text{MnO}_4^- \rightarrow \text{Mn}^{2+} (\text{aq})$
Oxidation half-reaction: $\text{Fe}^{2+} (\text{aq}) \rightarrow \text{Fe}^{3+} (\text{aq})$
- 4 Balance atoms other than H and O**
Mn and Fe are already balanced.
- 5 Balance oxygen using H_2O**
 $\text{MnO}_4^- (\text{aq}) \rightarrow \text{Mn}^{2+} (\text{aq}) + 4 \text{H}_2\text{O} (\text{l})$
- 8 Balance oxygen using electrons**
 $\text{MnO}_4^- (\text{aq}) + 8 \text{H}^+ (\text{aq}) \rightarrow 5 \text{Fe}^{2+} (\text{aq}) + 5 \text{e}^-$
- 9 Add half-reactions and cancel electrons**
 $\text{MnO}_4^- (\text{aq}) + 8 \text{H}^+ (\text{aq}) + 5 \text{e}^- \rightarrow \text{Mn}^{2+} (\text{aq}) + 4 \text{H}_2\text{O} (\text{l}) + 5 \text{Fe}^{2+} (\text{aq})$
- 8 Equalize electrons**
 $\text{MnO}_4^- (\text{aq}) + 8 \text{H}^+ (\text{aq}) \rightarrow 5 \text{e}^{2+} (\text{aq}) + 5 \text{e}^-$
- 10 Add half-reactions and cancel electrons**
 $\text{MnO}_4^- (\text{aq}) + 8 \text{H}^+ (\text{aq}) + 5 \text{Fe}^{2+} (\text{aq}) \rightarrow \text{Mn}^{2+} (\text{aq}) + 4 \text{H}_2\text{O} (\text{l}) + 5 \text{Fe}^{2+} (\text{aq})$
- 10 Final check**
Atoms balanced: Mn, Fe, O, and H
Charge balanced: Left side = +2, Right side = +2

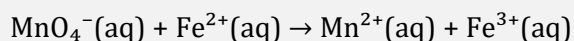
Example: Balance the following redox reaction in acidic solution:



This reaction occurs in acidic solution and is commonly used in redox titrations involving potassium permanganate.

Step 1 — Write the skeleton equation

The skeleton equation is already given with correct formulas and charges:



Step 2 — Assign oxidation numbers

- Mn in MnO_4^- has an oxidation number of +7.
- Mn^{2+} has an oxidation number of +2.

- Fe^{2+} has an oxidation number of +2.
- Fe^{3+} has an oxidation number of +3.

Manganese is reduced (+7 $\rightarrow$ +2).

Iron is oxidized (+2 $\rightarrow$ +3).

Step 3 — Write half-reactions

Reduction half-reaction: $\text{MnO}_4^-(\text{aq}) \rightarrow \text{Mn}^{2+}(\text{aq})$

Oxidation half-reaction: $\text{Fe}^{2+}(\text{aq}) \rightarrow \text{Fe}^{3+}(\text{aq})$

Step 4 — Balance atoms other than H and O

- Mn is already balanced.
- Fe is already balanced.

Step 5 — Balance oxygen using H_2O

The reduction half-reaction contains four oxygen atoms on the left.

Add 4 $\text{H}_2\text{O}(\text{l})$ to the right: $\text{MnO}_4^-(\text{aq}) \rightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\text{l})$

Step 6 — Balance hydrogen using H^+

There are 8 hydrogen atoms on the right, so add 8 $\text{H}^+(\text{aq})$ to the left:

$\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) \rightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\text{l})$

Step 7 — Balance charge using electrons

Left side charge: $(-1) + (+8) = +7$

Right side charge: +2

Add 5 e^- to the left to reduce the charge to +2: $\text{MnO}_4^-(\text{aq}) + 8 \text{H}^+(\text{aq}) + 5 \text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4 \text{H}_2\text{O}(\text{l})$

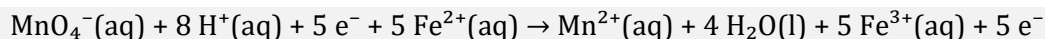
Oxidation half-reaction charge balance: $\text{Fe}^{2+}(\text{aq}) \rightarrow \text{Fe}^{3+}(\text{aq}) + \text{e}^-$

Step 8 — Equalize electrons

The reduction half-reaction uses 5 electrons.

Multiply the oxidation half-reaction by 5: $5 \text{Fe}^{2+}(\text{aq}) \rightarrow 5 \text{Fe}^{3+}(\text{aq}) + 5 \text{e}^-$

Step 9 — Add half-reactions and cancel electrons



Step 10 — Final check

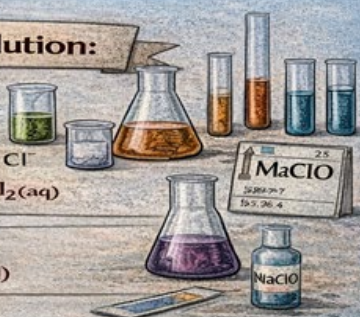
- Atoms balanced: Mn, Fe, O, and H
- Charge balanced: Left side = +2, Right side = +2

This equation is fully balanced for acidic solution.

PART C — Balancing Redox Equations in Neutral Solution (Worked Example)

Many redox reactions occur in water that is not strongly acidic. In neutral solution, you may still use $\text{H}_2\text{O}(\text{l})$ to balance oxygen, but you must ensure that $\text{H}^+(\text{aq})$ does not appear in the final balanced equation. In this part, you will balance a redox reaction in neutral solution using the half-reaction method and then remove H^+ correctly.

Example: Balance the following redox reaction in neutral solution:

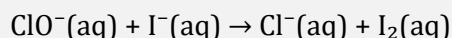
$$\text{ClO}^-(\text{aq}) + \text{I}^-(\text{aq}) \rightarrow \text{Cl}^-(\text{aq}) + \text{I}_2(\text{aq})$$


- Write the skeleton equation**
 $\text{ClO}^-(\text{aq}) + \text{I}^-(\text{aq}) \rightarrow \text{Cl}^-(\text{aq}) + \text{I}_2(\text{aq})$
- Assign oxidation numbers**
 ClO^- , oxygen is -2, so chlorine must be +1.
 I^- , in iodine is -1.
 Chlorine is reduced (+1 → -1).
- Balance atoms other than H and O**
 Oxidation half-reaction: $\text{ClO}^-(\text{aq}) \rightarrow \text{I}_2(\text{aq})$
- Balance oxygen using H_2O**
 $\text{ClO}^-(\text{aq}) \rightarrow \text{Cl}^-(\text{aq}) \rightarrow \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- Balance atoms other than H and O**
 Oxidation half-reaction:
 Balance iodine atoms by place in front of 2 front of I^- .
- Balance charge using electrons**
 Reduction half-reaction:
 Cl is already balanced.
 $\text{ClO}^-(\text{aq}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$
- Write half-reactions**
 Reduction half-reaction: $\text{ClO}^- \rightarrow \text{Cl}^-$
 Oxidation half-reaction: $\text{I}^-(\text{aq}) \rightarrow \text{I}_2(\text{aq})$
- Balance oxygen using H_2O**
 $\text{ClO}^-(\text{aq}) \rightarrow \text{Cl}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$
 Reduction half-reaction:
 Cl is already balanced.
- Equalize electrons and add half-reactions**
 $\text{ClO}^-(\text{aq}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{I}_2(\text{aq}) + 2 \text{OH}^-$
- Final check**
 Atoms balanced: Cl, I, O, and H.
 Charge check: Left side = -3, Right side = -3

PART D – Building Half-Reactions from Redox Information
 In many redox problems, you are not given a complete chemical equation. Instead, you may be provided with oxidation numbers, redox tables, or a description of which species are oxidized and reduced. In this part, you will learn how to construct correct half-reactions from this information.

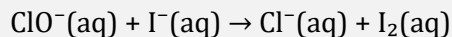
In the next part, you will apply half-reactions directly to quantitative redox.

Example: Balance the following redox reaction in neutral solution:



This type of reaction is important in real-world chemistry because hypochlorite (ClO^-) is the active oxidizing agent in household bleach, and iodide is used in analytical chemistry.

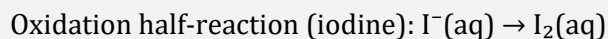
Step 1 — Write the skeleton equation



Step 2 — Assign oxidation numbers

- In ClO^- , oxygen is -2 , so chlorine must be $+1$.
- In Cl^- , chlorine is -1 . Chlorine is reduced ($+1 \rightarrow -1$).
- In I^- , iodine is -1 .
- In I_2 , iodine is 0 . Iodine is oxidized ($-1 \rightarrow 0$).

Step 3 — Write half-reactions



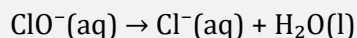
Step 4 — Balance atoms other than H and O



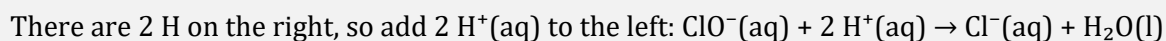
Step 5 — Balance oxygen using H_2O

The reduction half-reaction has one oxygen on the left.

Add 1 $\text{H}_2\text{O}(\text{l})$ to the right:



Step 6 — Balance hydrogen (temporary use of H^+)



Step 7 — Balance charge using electrons

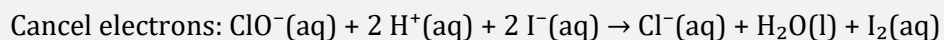
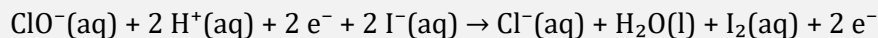
Reduction half-reaction charge: Left side charge = $(-1) + (+2) = +1$, Right side charge = -1

Add 2 e^- to the left to reduce the charge by 2: $\text{ClO}^-(\text{aq}) + 2 \text{H}^+(\text{aq}) + 2 \text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$

Oxidation half-reaction charge balance: $2 \text{I}^{-}(\text{aq}) \rightarrow \text{I}_2(\text{aq}) + 2 \text{e}^{-}$

Step 8 — Equalize electrons and add half-reactions

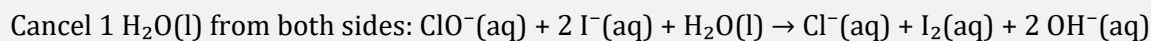
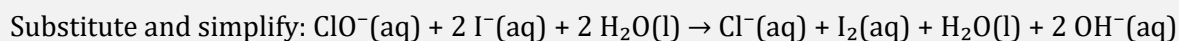
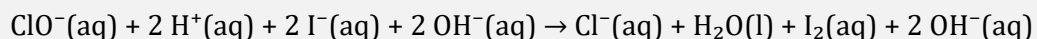
Both half-reactions involve 2e^{-} , so they can be added directly:



Step 9 — Convert to neutral conditions (remove H^{+})

H^{+} cannot appear in the final equation for neutral solution.

To eliminate $2 \text{H}^{+}(\text{aq})$, add $2 \text{OH}^{-}(\text{aq})$ to BOTH sides:



Step 10 — Final check

- Atoms balanced: Cl, I, O, H
- Charge check: Left side charge = $(-1) + 2(-1) + 0 = -3$, Right side charge = $(-1) + 0 + 2(-1) = -3$

This equation is fully balanced for neutral solution.

PART D — Building Half-Reactions from Redox Information

In many redox problems, you are not given a complete chemical equation. Instead, you may be provided with oxidation numbers, redox tables, or a description of which species are oxidized and reduced. In this part, you will learn how to construct correct half-reactions from this information.



Why Half-Reactions are Useful

Half-reactions isolate oxidation and reduction as separate processes. This allows chemists to focus on electron transfer directly and to build balanced equations even when only partial information is provided.

General strategy for building a half-reaction

Use the following steps whenever redox information is given:

- 1 Identify the species undergoing change.**
 - ✓ Determine which species is oxidized (loses electrons) and which is reduced (gains electrons).
- 3 Write the unbalanced half-reaction.**
 - ✓ Include only the species that changes oxidation number.
- 4 Balance atoms other than H and O.**
 - ✓ Oxidation half-reaction $\rightarrow \text{ClO}^-_{(\text{aq})} \rightarrow \text{Cl}^-_{(\text{aq})}$

General strategy for building a half-reaction

Use the following steps whenever redox information is given:

- ✓ Half-reactions must always include electrons.
- ✓ Oxidation half-reactions produce electrons.
- ✓ Reduction half-reactions consume electrons.

In the next part, you will apply half-reactions directly to quantitative redox titrations.

Example 1 — Constructing a half-reaction from oxidation numbers

Information given: Chromium changes from +6 in $\text{Cr}_2\text{O}_7^{2-}$ to +3 in acidic solution.

Step 1 — Identify the species: $\text{Cr}_2\text{O}_7^{2-} \rightarrow \text{Cr}^{3+}$

✓ Balance chromium atoms: $\text{Cr}_2\text{O}_7^{2-} \rightarrow 2 \text{Cr}^{3+}$

Step 3: Balance oxygen using H_2O :

✓ Add 7 H_2O to the right: $\text{Cr}_2\text{O}_7^{2-}$

Example 2 — Constructing a half-reaction from oxidation numbers

Information given: Chromium changes from +6 in $\text{Cr}_2\text{O}_7^{2-}$ to +3 in Cr^{3+} in acidic solution.

Step 1 — Identify the species:

✓ $\text{Cr}_2\text{O}_7^{2-} \rightarrow \text{Cr}^{3+} \rightarrow \text{Cr}^{3+} \rightarrow 2 \text{Cr}^{3+}$

Step 2: Balance chromium using H^+ :

✓ $\text{Cr}_2\text{O}_7^{2-} + 14 \text{H}^+ + 6 \text{e}^- \rightarrow 2 \text{Cr}^{3+} + 7 \text{H}_2\text{O}(\text{l}) + 2 \text{OH}^-$

Why half-reactions are useful

Half-reactions isolate oxidation and reduction as separate processes. This allows chemists to focus on electron transfer directly and to build balanced equations even when only partial information is provided.

A half-reaction must always show:

- the species undergoing change
- electrons (e^-) explicitly
- correct atom balance
- correct charge balance

General strategy for building a half-reaction

Use the following steps whenever redox information is given:

Step 1 — Identify the species undergoing change.

Determine which species is oxidized (loses electrons) and which is reduced (gains electrons).

Step 2 — Write the unbalanced half-reaction.

Include only the species that changes oxidation number.

Step 3 — Balance atoms other than H and O.**Step 4 — Balance oxygen using H₂O.****Step 5 — Balance hydrogen.**

- Use H⁺ in acidic solution.
- Use temporary H⁺ in neutral solution, then remove it later.

Step 6 — Balance charge using electrons.

Add e⁻ to the more positive side until total charge is equal.

Example 1 — Constructing a half-reaction from oxidation numbers

Information given: Chromium changes from +6 in Cr₂O₇²⁻ to +3 in Cr³⁺ in acidic solution.

Step 1 — Identify the species: Cr₂O₇²⁻ → Cr³⁺

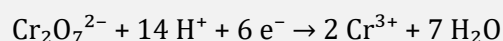
Step 2 — Balance chromium atoms: Cr₂O₇²⁻ → 2 Cr³⁺

Step 3 — Balance oxygen using H₂O: Add 7 H₂O to the right: Cr₂O₇²⁻ → 2 Cr³⁺ + 7 H₂O

Step 4 — Balance hydrogen using H⁺: Add 14 H⁺ to the left: Cr₂O₇²⁻ + 14 H⁺ → 2 Cr³⁺ + 7 H₂O

Step 5 — Balance charge using electrons: Left side charge = (-2) + (+14) = +12 Right side charge = 2(+3) = +6

Add 6 e⁻ to the left:



This is the correctly balanced reduction half-reaction.

Example 2 — Constructing a half-reaction from a redox table entry

Information given: Fe³⁺/Fe²⁺ is a redox couple.

Reduction half-reaction: Fe³⁺ + e⁻ → Fe²⁺

Oxidation half-reaction (reverse): Fe²⁺ → Fe³⁺ + e⁻

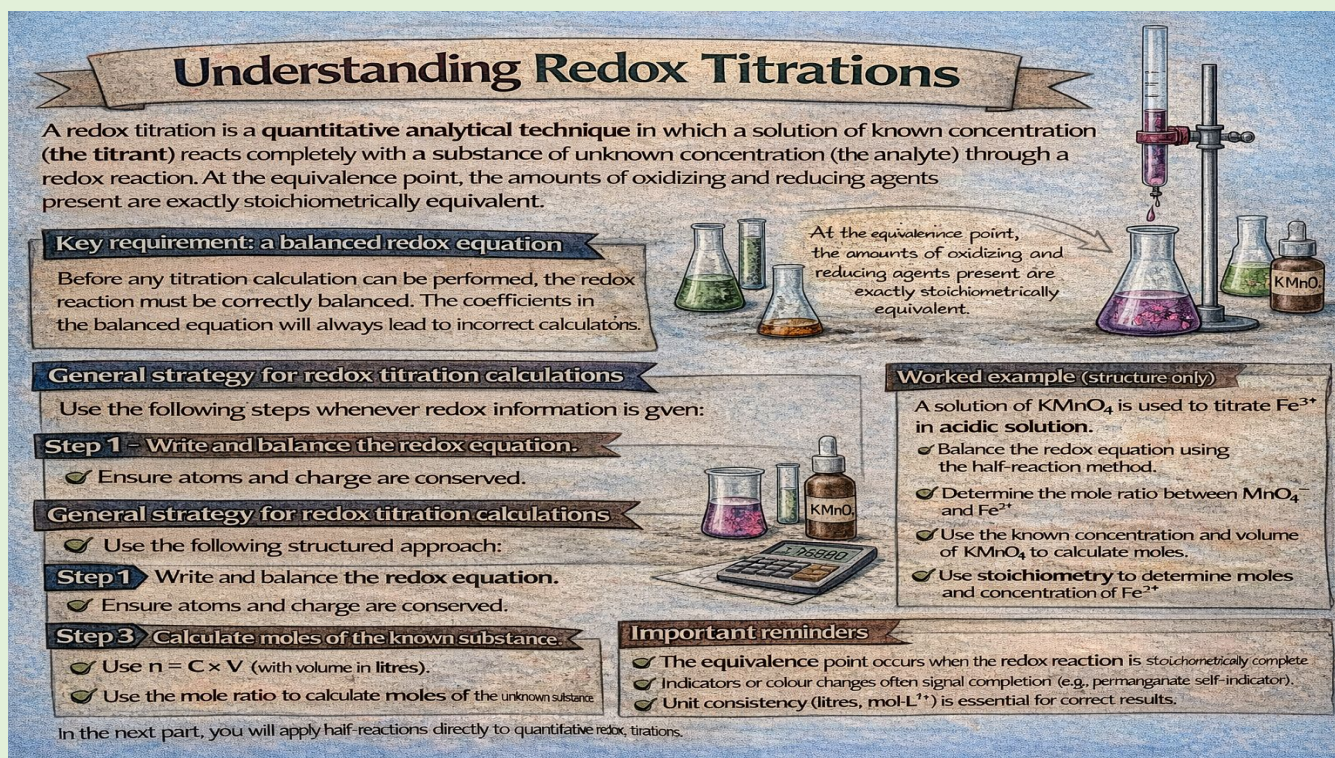
Important reminders

- Half-reactions must always include electrons.
- Oxidation half-reactions produce electrons.
- Reduction half-reactions consume electrons.
- Coefficients may be required to match electron transfer when half-reactions are combined.

In the next part, you will apply half-reactions directly to quantitative redox titrations.

PART E — Redox Titrations

Balanced redox equations are not only descriptive tools — they allow chemists to perform quantitative analysis. In this part, you will learn how redox reactions are used in titrations to determine unknown concentrations with precision.



Understanding Redox Titrations

A redox titration is a **quantitative analytical technique** in which a solution of known concentration (**the titrant**) reacts completely with a substance of unknown concentration (**the analyte**) through a redox reaction. At the equivalence point, the amounts of oxidizing and reducing agents present are exactly stoichiometrically equivalent.

Key requirement: a balanced redox equation

Before any titration calculation can be performed, the redox reaction must be correctly balanced. The coefficients in the balanced equation will always lead to incorrect calculations.

General strategy for redox titration calculations

Use the following steps whenever redox information is given:

Step 1 - Write and balance the redox equation.

- ✓ Ensure atoms and charge are conserved.

General strategy for redox titration calculations

- ✓ Use the following structured approach:

Step 1 Write and balance the redox equation.

- ✓ Ensure atoms and charge are conserved.

Step 3 Calculate moles of the known substance.

- ✓ Use $n = C \times V$ (with volume in litres).
- ✓ Use the mole ratio to calculate moles of the unknown substance.

At the equivalence point, the amounts of oxidizing and reducing agents present are exactly stoichiometrically equivalent.

Worked example (structure only)

A solution of KMnO_4 is used to titrate Fe^{3+} in acidic solution.

- ✓ Balance the redox equation using the half-reaction method.
- ✓ Determine the mole ratio between MnO_4^- and Fe^{3+} .
- ✓ Use the known concentration and volume of KMnO_4 to calculate moles.
- ✓ Use **stoichiometry** to determine moles and concentration of Fe^{3+} .

Important reminders

- ✓ The equivalence point occurs when the redox reaction is stoichiometrically complete.
- ✓ Indicators or colour changes often signal completion (e.g., permanganate self-indicator).
- ✓ Unit consistency (litres, mol-L^{-1}) is essential for correct results.

In the next part, you will apply half-reactions directly to quantitative redox titrations.

What is a redox titration?

A redox titration is a quantitative analytical technique in which a solution of known concentration (the titrant) reacts completely with a substance of unknown concentration (the analyte) through a redox reaction. At the equivalence point, the amounts of oxidizing and reducing agents present are exactly stoichiometrically

equivalent.

Redox titrations are widely used in chemistry, medicine, and industry. Examples include:

- determining iron(II) concentration in water samples
- analyzing hydrogen peroxide content in disinfectants
- measuring oxidizing strength of bleaches

Key requirement: a balanced redox equation

Before any titration calculation can be performed, the redox reaction must be correctly balanced. The coefficients in the balanced equation determine the mole ratios between reactants. An incorrect balanced equation will always lead to incorrect calculations.

General strategy for redox titration calculations

Use the following structured approach:

Step 1 — Write and balance the redox equation.

Ensure atoms and charge are conserved.

Step 2 — Identify the mole ratio.

Use coefficients from the balanced equation to relate oxidizing and reducing agents.

Step 3 — Calculate moles of the known substance.

Use $n = C \times V$ (with volume in litres).

Step 4 — Use the mole ratio to calculate moles of the unknown substance.

Step 5 — Calculate the unknown concentration or mass.

Worked example (structure only)

A solution of KMnO_4 is used to titrate Fe^{2+} in acidic solution.

- Balance the redox equation using the half-reaction method.
- Determine the mole ratio between MnO_4^- and Fe^{2+} .
- Use the known concentration and volume of KMnO_4 to calculate moles.
- Use stoichiometry to determine moles and concentration of Fe^{2+} .

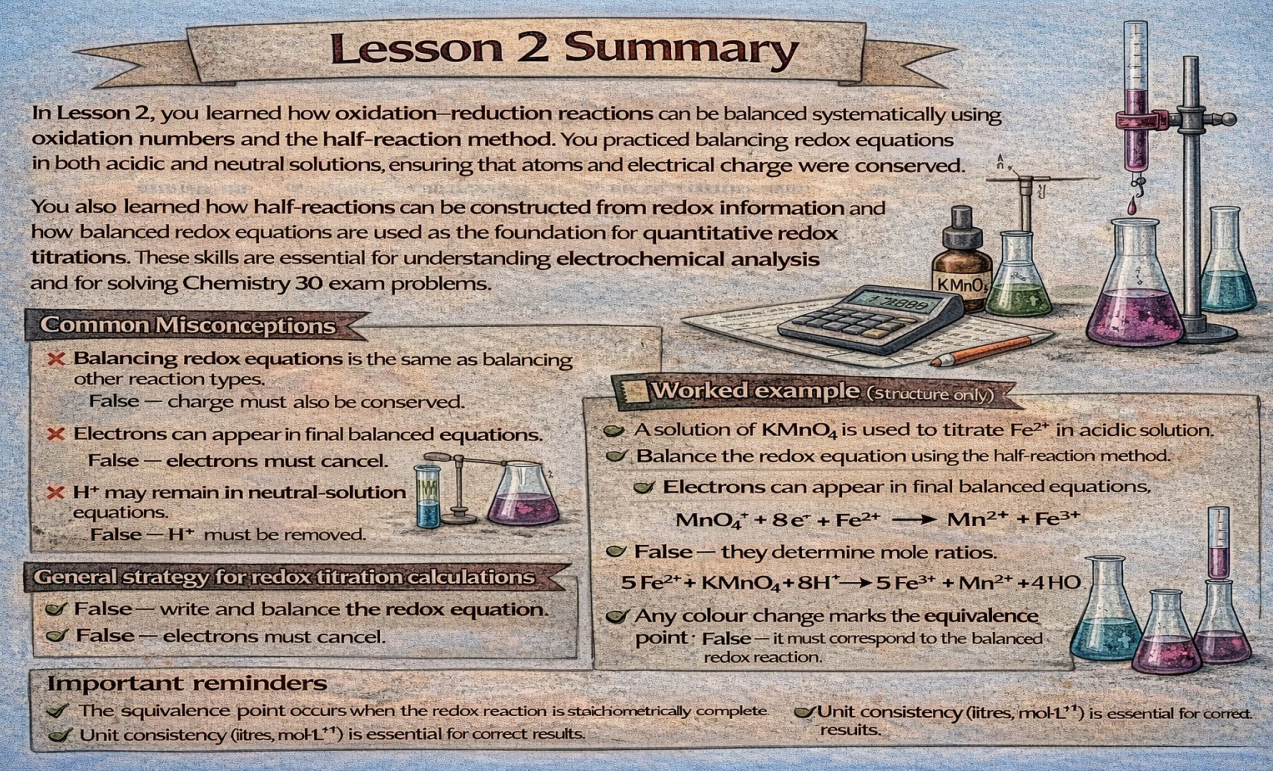
Detailed numerical examples will follow in practice questions and assignments.

Important reminders

- The equivalence point occurs when the redox reaction is stoichiometrically complete.
- Indicators or colour changes often signal completion (e.g., permanganate self-indicator).
- Unit consistency (litres, $\text{mol}\cdot\text{L}^{-1}$) is essential for correct results.

In the final section of this lesson, you will review key ideas and common errors before moving on to full problem-solving practice.

Lesson 2 Summary



Lesson 2 Summary

In Lesson 2, you learned how oxidation–reduction reactions can be balanced systematically using oxidation numbers and the half-reaction method. You practiced balancing redox equations in both acidic and neutral solutions, ensuring that atoms and electrical charge were conserved.

You also learned how half-reactions can be constructed from redox information and how balanced redox equations are used as the foundation for quantitative redox titrations. These skills are essential for understanding electrochemical analysis and for solving Chemistry 30 exam problems.

Common Misconceptions

- ✗ Balancing redox equations is the same as balancing other reaction types.
False — charge must also be conserved.
- ✗ Electrons can appear in final balanced equations.
False — electrons must cancel.
- ✗ H^+ may remain in neutral-solution equations.
False — H^+ must be removed.

Worked example (structure only)

- ✓ A solution of KMnO_4 is used to titrate Fe^{2+} in acidic solution.
- ✓ Balance the redox equation using the half-reaction method.
- ✓ Electrons can appear in final balanced equations.

$$\text{MnO}_4^- + 8\text{e}^- + \text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + \text{Fe}^{3+}$$

- ✓ False — they determine mole ratios.

$$5\text{Fe}^{2+} + \text{KMnO}_4 + 8\text{H}^+ \rightarrow 5\text{Fe}^{3+} + \text{Mn}^{2+} + 4\text{H}_2\text{O}$$

- ✓ Any colour change marks the equivalence point. False — it must correspond to the balanced redox reaction.

General strategy for redox titration calculations

- ✓ False — write and balance the redox equation.
- ✓ False — electrons must cancel.

Important reminders

- ✓ The equivalence point occurs when the redox reaction is stoichiometrically complete.
- ✓ Unit consistency (litres, $\text{mol}\cdot\text{L}^{-1}$) is essential for correct results.
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Common Misconceptions

- Balancing redox equations is the same as balancing other reaction types. (False — charge must also be conserved.)
- Electrons can appear in final balanced equations. (False — electrons must cancel.)
- H^+ may remain in neutral-solution equations. (False — H^+ must be removed.)
- The coefficients in a redox equation do not matter for calculations. (False — they determine mole ratios.)
- Any colour change marks the equivalence point. (False — it must correspond to the balanced redox reaction.)

Looking Ahead to Lesson 3 — Electrochemical Cells

In the next lesson, you will apply balanced redox equations to electrochemical cells. You will learn how oxidation and reduction occur at separate electrodes, how electrons flow through an external circuit, and how electrical energy is produced from chemical reactions.

The balancing skills you developed in Lesson 2 are essential for writing half-reactions at electrodes and for predicting cell behaviour.

Review Questions — Lesson 2

1. Balance the following redox equation in acidic solution: $\text{Cr}_2\text{O}_7^{2-} + \text{Fe}^{2+} \rightarrow \text{Cr}^{3+} + \text{Fe}^{3+}$

Solution: $\text{Cr}_2\text{O}_7^{2-} + 14 \text{H}^+ + 6 \text{Fe}^{2+} \rightarrow 2 \text{Cr}^{3+} + 7 \text{H}_2\text{O} + 6 \text{Fe}^{3+}$

2. Balance the following redox equation in neutral solution: $\text{ClO}^- + \text{I}^- \rightarrow \text{Cl}^- + \text{I}_2$

Solution: $\text{ClO}^- + 2 \text{I}^- + \text{H}_2\text{O} \rightarrow \text{Cl}^- + \text{I}_2 + 2 \text{OH}^-$

3. Write the reduction half-reaction for $\text{MnO}_4^- \rightarrow \text{Mn}^{2+}$ in acidic solution.

Solution: $\text{MnO}_4^- + 8 \text{H}^+ + 5 \text{e}^- \rightarrow \text{Mn}^{2+} + 4 \text{H}_2\text{O}$

4. A solution of KMnO_4 ($0.0200 \text{ mol}\cdot\text{L}^{-1}$) is used to titrate 25.0 mL of Fe^{2+} solution. If 18.6 mL of KMnO_4 is required, calculate the concentration of Fe^{2+} .

Solution:

$$n(\text{MnO}_4^-) = 0.0200 \times 0.0186 = 3.72 \times 10^{-4} \text{ mol}$$

$$\text{Mole ratio } \text{Fe}^{2+} : \text{MnO}_4^- = 5 : 1$$

$$n(\text{Fe}^{2+}) = 5 \times 3.72 \times 10^{-4} = 1.86 \times 10^{-3} \text{ mol}$$

$$C = n / V = 1.86 \times 10^{-3} / 0.0250 = 0.0744 \text{ mol}\cdot\text{L}^{-1}$$

5. Explain why a correctly balanced redox equation is essential before performing titration calculations.

Solution:

The coefficients in a balanced redox equation define the mole ratios between reactants. If the equation is incorrect, the calculated amount of substance reacting will be wrong, leading to incorrect concentration results.